

■ EV Infrastructure Capstone Project

Comprehensive Data Wrangling and Modeling Summary

1■■■ Project Overview

This capstone project focuses on forecasting electric vehicle (EV) adoption and charger infrastructure needs across Washington State through 2050. Using publicly available datasets from the Department of Licensing, the U.S. Department of Energy, and Tesla's Supercharger map, the analysis models future charging requirements under both lower-bound (geographic coverage) and upper-bound (population-based) scenarios.

2■■■ Datasets and Inputs

The analysis uses three core datasets: 1. Tesla Supercharger Network (`supercharger_in_washington_state.xls`) – Location, address, county, and GPS coordinates of existing chargers. 2. Population Dataset (`Population_2024_age_25_to_59.xlsx`) – County-level population (ages 25–59) from Washington OFM and U.S. Census. 3. EV Registration Dataset (`coordinates_output.xlsm`) – VIN-level anonymized BEV registrations with geographic coordinates.

3■■■ Data Wrangling Steps

The following cleaning steps were implemented across all scripts (King, Pierce, Kitsap, Chelan counties):

- Imported Excel data with pandas (`read_excel()`), converting object to numeric types.
- Standardized column names (e.g., County, State, total).
- Filtered records by state and county (e.g., "WA", "King County").
- Aggregated population, EV, and charger counts using `groupby()` and `len()`.
- Merged results to create a unified county-level dataset.
- Computed per-capita metrics: `EV_per_1000_residents` and `Chargers_per_1000_residents`.
- Saved results and plots to Excel using `openpyxl` and `matplotlib`.

4■■■ Handling Missing Values

Counties without chargers or EV registrations were assigned zeros (`fillna(0)`), ensuring no data loss. Missing coordinates were ignored in spatial analyses but retained for counts. Numeric fields were converted to int/float types to prevent calculation errors. No data rows were dropped to maintain statewide completeness.

5■■■ Outlier Management

King County displayed exceptionally high EV density (~102,000 VINs), while Chelan had very low counts. These were preserved, reflecting real-world demographic and infrastructure differences. YData Profiling confirmed that outliers were valid and should remain unaltered for modeling integrity.

6■■■ Modeling and Forecasting Approach

Two forecasting scenarios were applied for each county:

- **Lower Bound (Constant Build)** – Based on geographic area and charger coverage radius. Example: King County (2-mile radius) → ~184 chargers

needed for full coverage. • **Upper Bound (3-Phase Slowdown)** – Based on 1 charger per population benchmark (e.g., 1:1500–5000 ratio). Policy targets: • 45% EV adoption by 2030 • 70% by 2040 • 95% by 2050 • 80% residential charging by 2030 Each county projection was modeled from 2025–2050, calculating annual build rates and EV adoption curves.

7■■■ County-Level Model Results

County	Population (25–59)	Existing Chargers	Existing EVs	Lower Bound	Upper Bound (2050)
King	1,241,805	13	101,838	184	455
Pierce	448,201	1	16,277	64	93
Kitsap	125,820	0	6,428	8	22
Chelan	33,879	3	1,250	5	7

8■■■ Visualization Summary

Each model generated two key charts using Matplotlib: 1. Charger Projection Curve – compares constant build vs. 3-phase slowdown scenarios. 2. EV Adoption Rate – visualizes growth toward 45% (2030), 70% (2040), and 95% (2050) targets. Charts were embedded in Excel (``openpyxl.Image``) for reporting clarity and presentation.

9■■■ Tools & Libraries Used

- pandas – data loading, cleaning, merging
- numpy – numeric computation (coverage, ratios)
- matplotlib – visualization and time-series charting
- openpyxl – Excel writing and image embedding
- ydata_profiling – data profiling and validation
- reportlab – document generation (PDF export)

■ Conclusion

The completed data wrangling and modeling workflow provides a structured, data-driven framework for understanding EV adoption and charger infrastructure gaps in Washington State. The results offer a reproducible approach to estimating future charging requirements under multiple policy-driven scenarios, balancing population growth, geographic distribution, and clean energy mandates.